

Indian Summer Monsoon Response to Regional Aerosol Emission Changes

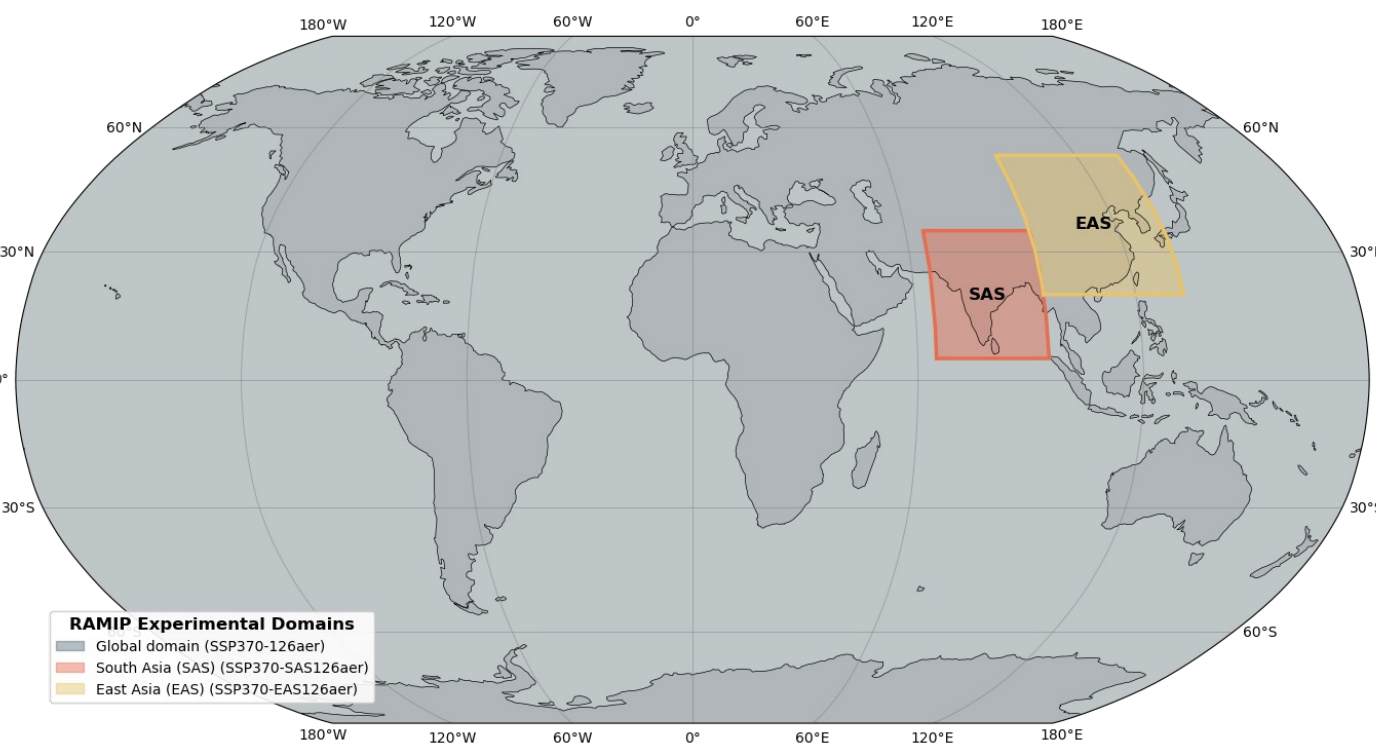


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Background

- The South Asian summer monsoon provides 80% of annual precipitation between JJAS (Fang et al., 2023).
- Over South Asia, changes in anthropogenic aerosol emissions have been a key driver of historical monsoon trends (Bollasina et al., 2011; Undorf et al., 2018).
- Exposure to PM linked to increased risks of respiratory and cardiovascular diseases (Rahman et al., 2022).
- Future projections under various SSPs anticipate substantial reductions in aerosol emissions due to air quality policies (Wilcox et al., 2023).



- SSP370-126aer:** Global aerosol emissions follow SSP1-2.6
- SSP370-EAS126aer:** East Asian aerosol emissions follow SSP1-2.6
- SSP370-SAS126aer:** South Asian aerosol emissions follow SSP1-2.6

Methodology

Regional Aerosol Model Intercomparison Project (RAMIP) Transient Experiments

RAMIP experiments are designed to quantify the role of regional aerosol emissions changes in near-term projections (Wilcox et al., 2023).

- GHGs, ozone, natural emissions from SSP3-7.0
- Anthropogenic aerosols include SO₂, SO₄, BC, OC
- 10 ensemble members each (2015-2050) for **CanESM5-1, CNRM-ESM2-1, EC-Earth3-AerChem, GFDL-SPEAR, MIROC6, MRI-ESM2-0, NorESM2-LM, UKESM1-0-LL**.

Ensemble Processing

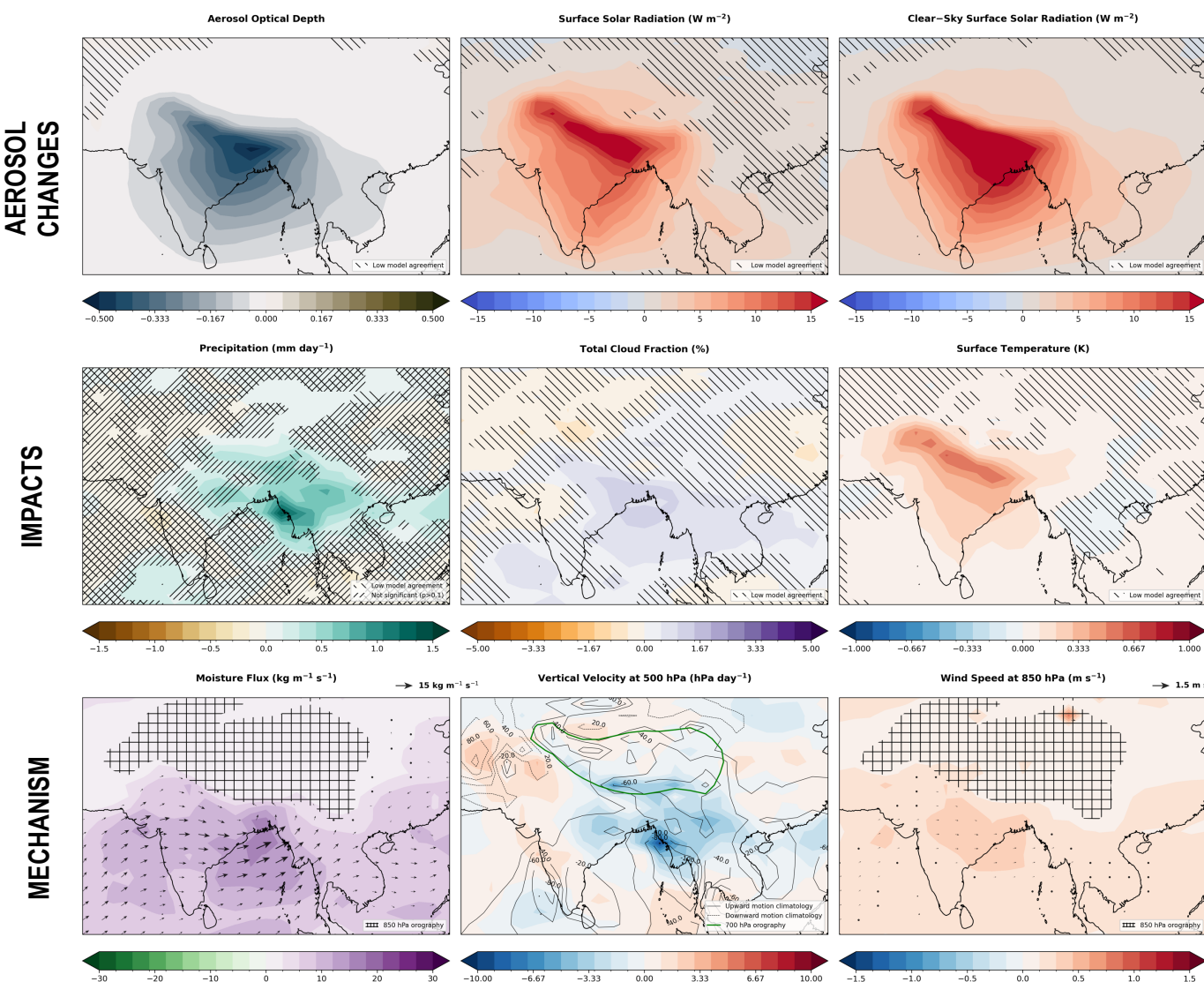
- 10 ensemble members per experiment
- All pairwise combinations: **exp[i] - base[j]**
- 10×10 = 100 difference combinations (Maximises statistical sampling)

Multi-model Mean Response

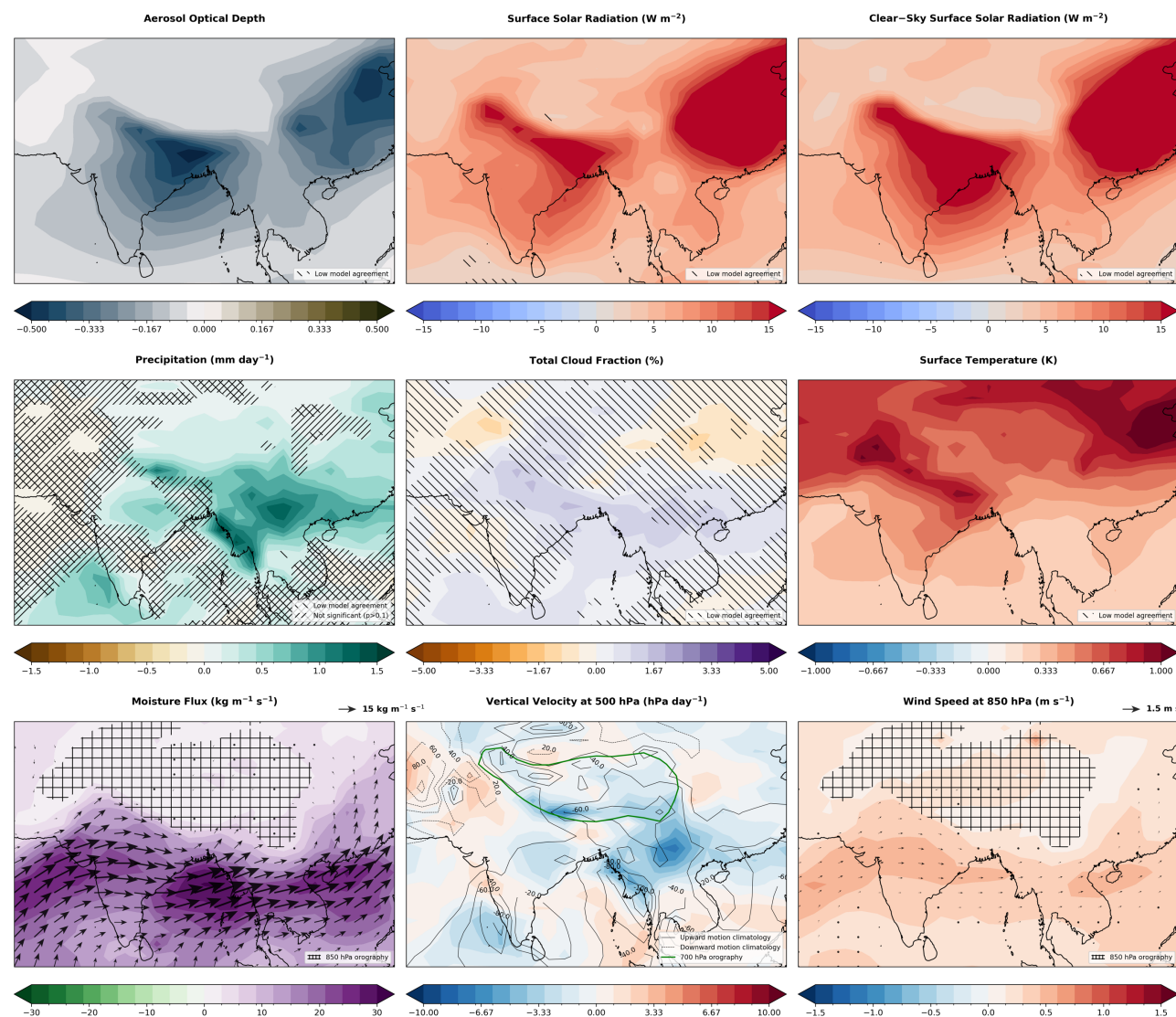
- Regrid all model output to common grid
- JJA seasonal averages (2041-2050)
- Calculate model-by-model differences
- Model agreement metrics (70% threshold)

Results

MMM response to South Asian aerosol reduction (2041-2050)



MMM response to Global aerosol reduction (2041-2050)



MMM response to East Asian aerosol reduction (2041-2050)

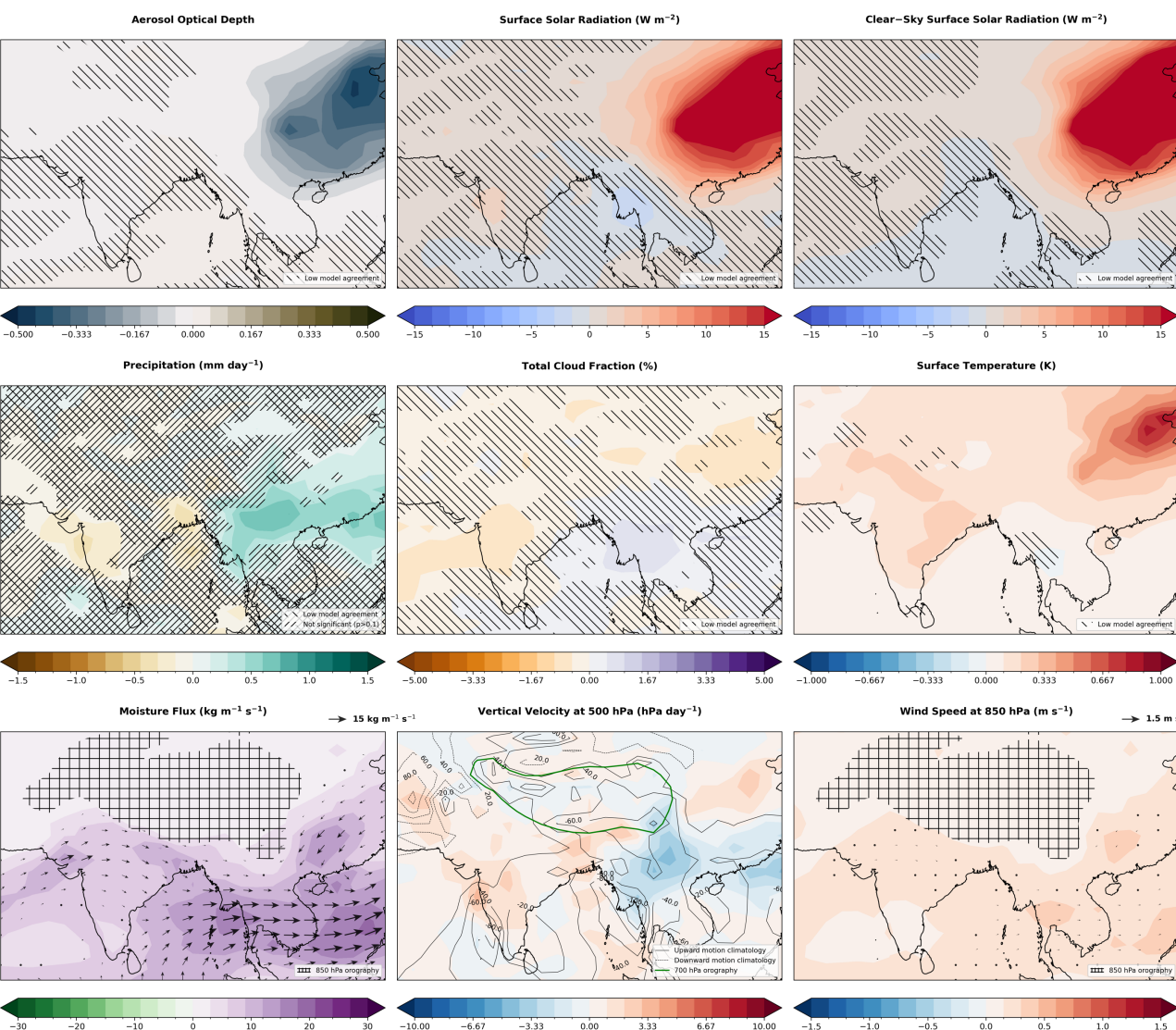


Figure. Multi-model mean response to South Asian aerosol reduction (ssp370-SAS126aer - ssp370), Global aerosol reduction (ssp370-126aer - ssp370) and East Asian aerosol reduction (ssp370-EAS126aer - ssp370) for 2041-2050: (a) Aerosol Optical Depth, (b) Surface Solar Radiation, (c) Clear-Sky Surface Solar Radiation, (d) Precipitation, (e) Total Cloud Fraction, (f) Surface Temperature, (g) Moisture Flux, (h) Vertical Velocity at 500 hPa, (i) Wind Speed at 850 hPa. Regions without backslashes (\\) show where at least 70% of models agree on sign of change for (a), (b), (c), (d), (e), (f). Regions without forward slashes (///) indicate areas where the changes are statistically significant at the 10% significance level regions for (d).

Summary

- South Asian aerosol reductions lead to strong local monsoon enhancement through direct radiative warming, increased atmospheric instability, and strengthened moisture transport.
- Global aerosol reductions amplify these effects and add interhemispheric temperature gradient changes that shift the ITCZ northward, further intensifying the Indian Summer Monsoon.
- East Asian aerosol reductions suppress the South Asian monsoon through atmospheric teleconnections, including: a seesaw effect where enhanced ascent over East Asia induces subsidence over South Asia, reduction in the thermal gradient driving monsoonal flow into India, diversion of moisture convergence toward East Asia.
- Coordinated approaches to aerosol emissions may be necessary to effectively manage future monsoon changes and their impacts.

References

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